Detailed Project Note: Insurance Fraud Detection with Machine Learning

1. What the Project Is About

This is a **Data Science coursework project (OCOM510M – Assessment 2, Student ID 202027438)** that builds a **predictive model to detect fraudulent insurance claims** for an insurance company. Using historical claim data, the project develops and compares **two machine learning classifiers** — **Logistic Regression** and **Random Forest** — to predict whether a customer/claim is fraudulent ("ReportedFraud" as the binary target: 0 = Legitimate, 1 = Fraud).

Core aim: Help the insurer identify potentially fraudulent claims in order to **(a) reduce financial losses from paid-out fraudulent claims** and **(b) avoid wrongly flagging legitimate customers** (which causes customer loss and reputational cost).

Key objectives:

1. **Data acquisition & preparation** – merge five source files (Demographics, Policy, Claim, Vehicle, Fraud target) into one dataset and handle missing values.
2. **Model development & justification** – prototype workflows in **RapidMiner**, then implement in **Python (scikit-learn)**.
3. **Performance evaluation & reporting** – cross-validation, confusion matrices, Balanced Accuracy, F1-score, Recall, ROC curves.
4. **Financial impact (cost–benefit) modelling** – build a pricing model and translate prediction errors (false positives/negatives) into monetary losses.

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2. Methodology Summary

Data & Preprocessing

- Merged datasets → 115,344 rows × 40 columns → duplicate CustomerIDs removed → **28,836 samples, 39 features**.

- Fraud distribution: **27% fraud (7,785) vs 73% non-fraud (21,051)** → imbalance ratio **2.70:1**, addressed with `class_weight='balanced'`.
- Handled 9 missing-value placeholder types; high-missingness fields included PropertyDamage (36.3%), PoliceReport (34%), TypeOfCollission (17.9%).
- Leak-safe imputation (median/most-frequent fit on training data only), 14 missing-indicator features, IQR outlier capping on claim amounts, label + one-hot encoding (~5,300 columns), StandardScaler, variance filtering, correlation removal, and **ANOVA SelectKBest reduction to 100 features**.
- Engineered features: PolicyDuration, ClaimDelay, ClaimToPremiumRatio, InjuryToTotalRatio, SeverityNumeric, SeverityXClaim.
- **Stratified 70/30 train-test split** (20,185 train / 8,651 test).

Models

- **Logistic Regression:** GridSearchCV (C=0.01, L2, liblinear, balanced weights) + **RFE feature selection (30 features)**.
- **Random Forest:** GridSearchCV over 162 candidates (best: 400 trees, depth 15, log2 features) + **top-30 importance-based features**.

Test-set performance

Metric	Logistic Regression	Random Forest
Accuracy	0.8108	**0.8532**
Balanced Accuracy	0.7911	**0.8323**
Precision	0.6250	**0.7042**
Recall	0.7483	**0.7868**
F1-Score	0.6811	**0.7432**
ROC-AUC	0.8405	**0.8977**
Nested CV Score	0.7944 ± 0.0030	**0.8592 ± 0.0031**

- **Overfitting check:** Train-test gaps of 0.0042 (LR) and 0.0298 (RF) → no significant overfitting; nested CV confirms both models are stable.
- **Fraud signals found:** missing severity/collision data, high total claim amounts, high umbrella limits, hobbies (chess, cross-fit) increase fraud risk; minor/trivial damage and consistent details reduce it.

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3. Relevance

- **Business relevance:** Insurance fraud is a multi-billion-dollar industry problem. The model converts abstract classification errors into **actionable monetary costs**, letting the client compare models on financial grounds, not just accuracy. Fraud flags also directly affect **customer retention** (false positives alienate genuine claimants) and **pricing strategy**.
- **Methodological relevance:** Demonstrates a full, industry-style ML pipeline — leak-free preprocessing, class-imbalance handling, hyperparameter tuning, **nested cross-validation** for unbiased estimates, overfitting diagnostics, and interpretability analysis (coefficients & feature importances).
- **Balanced decision-making:** The project explicitly weighs the trade-off between **detection power (recall)** and **false accusation risk (precision)**, mirroring real regulatory and ethical concerns in insurance.

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4. Financials

Pricing model (per project assumptions):

- Average claim amount: **$52,308.55**
- Claim rate: **10%**
- Gross profit multiplier: **2**
- **Annual policy price = (Avg claim × 2) ÷ claim rate = $1,046,170.91**
- Average fraudulent claim ≈ **$58,745** (derived from FN costs)

Cost of model errors (test set): False Positives cost the *annual policy price* (lost customers); False Negatives cost the *average fraud claim* (paid-out fraud).

Cost Item	Logistic Regression	Random Forest
False Positives	1,049	772
False Negatives	588	498
Cost of False Positives (CFP)	$1,097,433,284.03	$807,643,942.11
Cost of False Negatives (CFN)	$34,542,004.75	$29,254,963.21
TOTAL COST	**$1,131,975,288.78**	**$836,898,905.32**

Cost comparison:

- **Random Forest saves $295,076,383.47** per year versus Logistic Regression.
- **Recommended model: Random Forest** — it wins on *every* criterion (best Balanced Accuracy, F1, ROC-AUC, and lowest total cost), driven by both fewer false alarms and fewer missed frauds.

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5. Key Findings & Limitations

- **Recommendation:** Random Forest, for "lower cost with comparable/superior performance."
- **Target gap:** The client's target was a **5% error rate (95% balanced accuracy)**; Random Forest achieved **83.23% balanced accuracy (16.77% error)** — best available, but the target was **not met**.
- **Not accomplished / future work:** XGBoost/LightGBM boosting, cost-optimized decision thresholds, SHAP/LIME explanations, time-based validation, geographic/behavioural features, graph neural networks for fraud rings, and Bayesian optimisation to reduce the ~$1B false-positive exposure.
- **Self-critique:** lack of side-by-side fraud vs non-fraud EDA, reliance on static variables, narrow algorithmic scope, and tuning aligned to accuracy metrics rather than financial risk.

Bottom line: The project delivers a deployable fraud-detection pipeline whose standout contribution is its **financial framing** — showing that model choice translates directly into hundreds of millions of dollars in difference, with Random Forest as the clear winner.